Tribhuvan University

Faculty of Management

OFFICE OF THE DEAN

2024

Bachelor in Business Administration

Course Title: **Operations Management**

Code No: **MGT205** Semester: **5th Semester**

Full Marks: **60** Pass Marks: **24** Time: **a production is made. The production cost is Rs 2 per item and the inventory carrying cost is 50 paisa per unit per month. If the shortage cost is Rs 3 per item per month, determine how often to make a production run and of what size?Section "B"Short Answer Questions:(Attempt any SIX Questions)[6 × 5 = 30]11.Describe transformation process with appropriate example.12."Operations strategy acts as a competitive weapon." Justify this statement.13.Describe the emerging issues in product and service design.14.The annual demand of a product is 10000 units. Each unit costs Rs.100 if orders placed in quantities below 200 units but for orders of 200 or above the price is Rs. 95. The annual inventory holding costs is 10 percent of the value of the item and the ordering cost is Rs. 5 per order. Find the economic lot size.15.A businessman has three alternatives open to him and each of which can be followed by any of the four possible events. The conditional payoffs foe each action event combination are given below:ActionEventsABCDP180-106P2-41218-2P314998What action should be chosen, if the criterion of choice is i. Maximax ii. Hurwitz criterion if the coefficient of pessimism is 0.60? iii. Laplace Criterion16.Calculate value of the central line and the control limits for the mean chart and the range chart and then comment on the state of controlSample No.123456789Mean151715181714181517 16Range77498712411(Conversion factors for n = 5 is A2 = 0.58, D3 = 0, and D4 = 2.115)17.The ABC Motors has four machines on which to do four jobs. Each job can be assigned to only one machine. The cost of each job on each machine is given in the following table. What are the job assignments which will minimize costs?MachinesJobs1234W35114X2169Y31027Z10615Section "C"Long Answer Questions:(Attempt any THREE Questions)[3 × 10 = 30]18.Define the term quality control. Explain seven tools for quality.19.Determine the minimum transportation cost from the following matrix.FactoryStores (Haulage cost in Rs.)AvailabilityS1S2S3S4F19710814F281191127F31310121014Requirement151911105520.A retailer has to decide as to the optimum number of units to be stocked of certain items under the following condition.(a) Cost price in season is Rs 15(b) Bargain price after season is Rs 9(c) Cost of holding the item beyond season is Rs 2(d) Selling price in season is Rs 20The probability distribution of demand based on past data is as follows:Demanded units1213141516Probability0.200.200.250.200.15Determine the optimum stock level on the EMV criterion and the expected value.21.Reduce the following two-person zero sum game to 2x2 order by dominance rule and obtain the optimal strategies for each player and the value of the game.Player BPlayer AB1B2B3B4A13240A23424A30440A44408Section "D"Comprehensive Answer / Case / Situation Analysis Questions:[4 × 5 = 20]22.Hetauda Kapada Udhyog Ltd. is located at Hetauda, Nepal. The company is working in textile business activities. In the early phase, the company is actively producing and distributing its product in the Nepalese market. Its main domain is to produce sari in the Nepalese market. In the early stage of production innovative technology is acquired by the company to produce a bulk amount of Sari. Product design and product evaluation for a certain time period had been done in an effective way. Later on, work on product design is slower.A new executive director was appointed to the company to manage the operation. Trade unions are creating barriers to employee reward and punishment. Different strategies are used in product design that did not last on market. The developed product is in a testing phase where product quality is measured before the supply of the product. The quality of the product is not measured since its establishment. Last year the sales of the company were just Rs 5,000,000. Which is less in comparison to the previous year.The market position of the company is being deteriorated as per the marketing manager. The product is almost out of the market and later on, the company was bankrupt and the government need to close it. Questions:a. What are the major problems faced by the operations manager in this case?b. Mention how good product design helps to maintain the quality of the product.c. What are the different product design strategies that can be used to adopt the change?d. On behalf of the operations manager how could you cope with the challenges in the operational department?**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

Brief Answer Questions:[10 × 2 = 20]

1. Define continuous production system.

2. Highlight key issues for operations manager.

3. List out the advantages of operations strategy.

4. State the importance of game theory.

5. What is value analysis?

6. Write down the assumptions of single channel queuing system.

7. List out the stages of historical evolution of Total Quality Management.

8. Define dependent and independent demand.

9. Determine whether the given two-person zero sum game is strictly determinable and fair.Player BPlayer AB1B2A112A24-3

10. The demand of an item is uniform at a rate of 25 units per month. The fixed cost is Rs 30 each time a production is made. The production cost is Rs 2 per item and the inventory carrying cost is 50 paisa per unit per month. If the shortage cost is Rs 3 per item per month, determine how often to make a production run and of what size?

Section B

Short Answer Questions:(Attempt any SIX Questions)[6 × 5 = 30]

11. Describe transformation process with appropriate example.

12. "Operations strategy acts as a competitive weapon." Justify this statement.

13. Describe the emerging issues in product and service design.

14. The annual demand of a product is 10000 units. Each unit costs Rs.100 if orders placed in quantities below 200 units but for orders of 200 or above the price is Rs. 95. The annual inventory holding costs is 10 percent of the value of the item and the ordering cost is Rs. 5 per order. Find the economic lot size.

15. A businessman has three alternatives open to him and each of which can be followed by any of the four possible events. The conditional payoffs foe each action event combination are given below:ActionEventsABCDP180-106P2-41218-2P314998What action should be chosen, if the criterion of choice is i. Maximax ii. Hurwitz criterion if the coefficient of pessimism is 0.60? iii. Laplace Criterion

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Section C

Long Answer Questions:(Attempt any THREE Questions)[3 × 10 = 30]

18. Define the term quality control. Explain seven tools for quality.

19. Determine the minimum transportation cost from the following matrix.FactoryStores (Haulage cost in Rs.)AvailabilityS1S2S3S4F19710814F281191127F31310121014Requirement1519111055

20. A retailer has to decide as to the optimum number of units to be stocked of certain items under the following condition.(a) Cost price in season is Rs 15(b) Bargain price after season is Rs 9(c) Cost of holding the item beyond season is Rs 2(d) Selling price in season is Rs 20The probability distribution of demand based on past data is as follows:Demanded units1213141516Probability0.200.200.250.200.15Determine the optimum stock level on the EMV criterion and the expected value.

21. Reduce the following two-person zero sum game to 2x2 order by dominance rule and obtain the optimal strategies for each player and the value of the game.Player BPlayer AB1B2B3B4A13240A23424A30440A44408

Section D

Comprehensive Answer / Case / Situation Analysis Questions:[4 × 5 = 20]

22. Hetauda Kapada Udhyog Ltd. is located at Hetauda, Nepal. The company is working in textile business activities. In the early phase, the company is actively producing and distributing its product in the Nepalese market. Its main domain is to produce sari in the Nepalese market. In the early stage of production innovative technology is acquired by the company to produce a bulk amount of Sari. Product design and product evaluation for a certain time period had been done in an effective way. Later on, work on product design is slower.A new executive director was appointed to the company to manage the operation. Trade unions are creating barriers to employee reward and punishment. Different strategies are used in product design that did not last on market. The developed product is in a testing phase where product quality is measured before the supply of the product. The quality of the product is not measured since its establishment. Last year the sales of the company were just Rs 5,000,000. Which is less in comparison to the previous year.The market position of the company is being deteriorated as per the marketing manager. The product is almost out of the market and later on, the company was bankrupt and the government need to close it. Questions:a. What are the major problems faced by the operations manager in this case?b. Mention how good product design helps to maintain the quality of the product.c. What are the different product design strategies that can be used to adopt the change?d. On behalf of the operations manager how could you cope with the challenges in the operational department?